

A small adder provenance question

ASTERION | James C. Simmonds | 3 October 2026

I am seeking advice about one saved FTCircuitBench artifact, not claiming a new compilation technique or a defect in the current tool.

Pinned comparison

Repository commit **116df005d68404e4c21cbd91b570c171deb06f7c**. Original: qasm/adder/adder_10q.qasm. Saved compilation: circuit_outputs/adder_10q_gs_clifford_t.qasm. Preparation X operations are retained; only final measurements are removed. The original expands to five X, seventeen CNOT and eight Toffoli gates.

Saved adder artifact	Standard exact baseline
1,120 T	32 T + 24 T-dagger
65 CNOT	65 CNOT
1,202 H + 210 S	16 H + 5 X

The exact baseline uses the standard seven-T-family Toffoli identity, checked in the algebraic field $Q(\sqrt{2}, i)$. Its full-unitary maximum entrywise difference from the original is $3.78e-15$ numerically. This is a known baseline, not an optimality result.

Saved artifact numerical check

All 1,024 input columns were compared using one global phase from $\text{trace}(U_{\text{original}}^* U_{\text{compiled}})$, with the original adjointed. At that phase the operator residual is **0.2031076762** and the Frobenius residual is 1.5508449558. This is not an optimization of operator norm over phase. The configured zero-input correct output probability is **0.9998993047**. Worst error over all full-register basis inputs is 0.0038585430; that set includes states outside usual zeroed-ancilla arithmetic inputs.

Qiskit CPU and two independently applied complex-double NVIDIA replays agree to $1.05e-13$. A separately written MATLAB parser/unitary audit is included in the review package. Numerical metrics are not interval certificates. The paper explicitly describes initial unoptimized compilations; the 20:1 T-family ratio is not a physical-cost or speedup claim.

Question for the authors

Which synthesis tolerance and tool revision generated this saved adder, and is it intended mainly as an initial unoptimized baseline? Would a small automated check comparing known exact Clifford+T subcircuits and whole-unitary residuals be a useful contribution, or does an existing test already cover this?

FTCircuitBench paper | Pinned source. Source license retained. Independent operator-assisted review only; external acceptance pending.